



Systematic review and meta-analysis

Rhizosheath: Distinct features and environmental functions

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ARTICLE INFO

Handling Editor: Cornelia Rumpel

Keywords:

Environmental stress

Plant growth

Rhizosphere biogeochemistry

Root–soil interface

Sandy soil

ABSTRACT

Interactions between plant roots and the surrounding soils are crucial for plant growth and soil health. The rhizosheath, which is the sheath-like soil that is tightly attached to the plant root surface, is increasingly acknowledged to exhibit distinct features and critical environmental functions for soil, plants, and microbes. However, the differences between the terms “rhizosheath” and “rhizosphere soil” remain ambiguous across the literature, and not all features and functions of rhizosheaths have reached a consensus. Here, we summarize the terminology, methodology, formation mechanisms, and findings of the features and functions of rhizosheaths via an integrated and quantitative approach. First, we propose a comprehensive framework for the terminology of different rhizosphere compartments, which allows for consistent definitions and improved comparability across studies. We also summarize the current knowledge of the mechanisms underlying rhizosheath formation. Moreover, by collecting evidence from the literature, we highlight the distinct physical, chemical, and microbial features of rhizosheath soils, such as 66% higher water content, 34% higher soil organic carbon, 56% higher water-extractable organic carbon, and 71% higher microbial biomass nitrogen in rhizosheath soils than those in other rhizosphere counterparts. We further discuss how these features lead to the environmental and biogeochemical functions of rhizosheaths, including sand fixation, soil aggregation, water and nutrient uptake, carbon dynamics, and microbial colonization. We highlight the significance of these functions from micro- to ecosystem-scale. Finally, we propose future research perspectives to better understand and harness the environmental benefits of rhizosheaths.

1. Introduction

Rhizosphere is an active zone of the terrestrial ecosystem, where physical, chemical, and biological features are greatly affected by root–soil–microbe interactions (Bhaduri et al., 2015). Root–soil–microbe interactions play a critical role in regulating terrestrial ecosystem nutrient and carbon biogeochemical cycling (Hallett et al., 2022), with rhizodeposition being the key driver (Ndour et al., 2021). Rhizodeposition is the release of carbon compounds from living roots to the soil (Lambers et al., 2009), which accounts for 30–40% of root-derived organic carbon inputs to soil (Ludwig et al., 2007). Rhizodeposits are primarily released from the distal elongation zone and behind the root tip, where root hairs are often thick and tight (Marschner et al., 2011). Certain environmental circumstances, including drought, nutrient

deficiency, root damage, and elevated levels of atmospheric carbon dioxide, are likely to stimulate plants to allocate more photosynthates into rhizosphere via enhanced rhizodeposition (Nguyen, 2003; Albalasmeh and Ghezzehei, 2014). Strong root–soil–microbe interactions can occur in the presence of thick and tight root hairs, massive rhizodeposits, and active microbes, which facilitate the soil aggregation around the root systems, therefore forming a sheath-like structure on the root surface.

This special morphological and physiological trait is referred to as rhizosheath (Pang et al., 2017) or as sandy sheath in early literature (Duell & Peacock, 1985; Price, 1911), which is considered a typical manifestation of plants' adaptive evolution (Ndour et al., 2020). Rhizosheath is a specific type of aggregated soil attached to the root systems and differs significantly from the loosely attached rhizosphere soil and bulk soil. Morphologically, the thickness of rhizosheaths ranges from 0.3

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<https://doi.org/10.1016/j.geoderma.2023.116500>

Received 25 January 2023; Received in revised form 14 April 2023; Accepted 4 May 2023

Available online 12 May 2023

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to 4.2 mm and shows great variations across plant species (North and Nobel, 1997; Qiu et al., 2012; Hartnett et al., 2013). For example, Hartnett et al. (2013) observed that rhizosheath thickness was 0.3 mm for non-drought-tolerant grasses and 0.5 mm for drought-tolerant grasses. Technically, rhizosheath soils can stay tightly attached to the fresh roots following excavation and hand-shaking (Brown et al., 2017). Early in 1887, rhizosheaths were first discovered in the roots of desert plants and subsequently attracted widespread attention. In recent years, an increasing number of studies have revealed that rhizosheath is a promising adaptive root–soil interface trait that provides essential functions (Ndour et al., 2020; Hallett et al., 2022), including sand fixation (Othman et al., 2004), plant drought resistance (Basirat et al., 2019), and plant water and nutrient acquisition (Rabbi et al., 2021). These environmental functions of rhizosheaths are critical for plant growth under environmental stress and assist plants to better cope with harsh habitats under water- and nutrient-deficient conditions.

Moreover, it is reported that rhizosheaths are present in many angiosperm species, especially in herbaceous and shallow-rooted plants (Brown et al., 2017). The ubiquitous nature of rhizosheaths suggests their significance to plant growth and rhizosphere processes (Burak et al., 2021). Increasing field observations show that under similar environmental conditions or within one specific ecosystem, some

species are more likely to form strong rhizosheaths, while others do not (Brown et al., 2017; Mo et al., 2022a; Tian et al., 2022). Current studies have recorded that most of the *Poales* and some of the *Fabales*, *Caryophyllales*, *Commelinales*, and *Lamiales* usually have clear rhizosheath structures, while most of the *Chrysanthemum*, *Zygophyllales*, *Malvales*, and some of the *Brassicaceae* are usually lack of rhizosheaths (Bailey and Scholes, 1997; Smith et al., 2011; Brown et al., 2017). Therefore, we propose that the presence and absence of rhizosheaths in plant roots can be considered two different situations. According to the occurrence of rhizosheaths in field observations, plant species can be subjected as rhizosheath-forming (RF) when a clear rhizosheath structure is presented; otherwise, it can be subjected as non-rhizosheath-forming (NRF). Fig. 1 demonstrates the typical structure of rhizosheaths formed in various species and clearly shows the morphological differences between root systems with and without rhizosheaths. Noted, although some species are widely reported to have rhizosheaths, which can be considered typical RF plants, the occurrence and strength of rhizosheaths are also affected by soil characteristics (e.g., texture, mineralogy) and environmental conditions. There are some species that lie in the between with varied rhizosheath scores (Brown et al., 2017), and some typical RF plants can sometimes lack rhizosheaths due to the growing conditions. For example, rice usually has a large rhizosheath

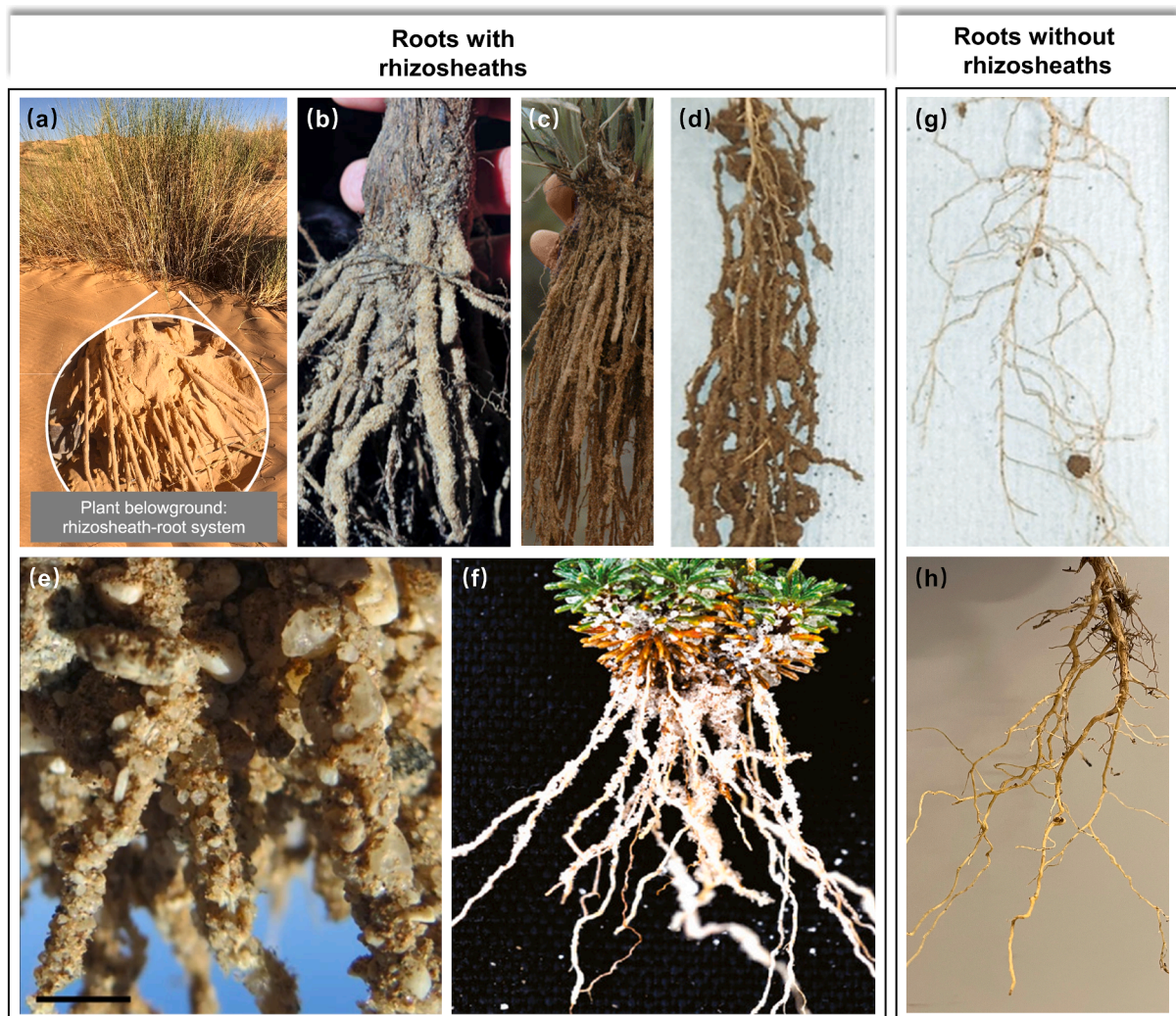


Fig. 1. Images of different plant roots with and without rhizosheaths. (a) *Stipagrostis pungens* (Marasco et al., 2022a), (b) *Haemodorum paniculatum* (Smith et al., 2011), (c) *Conostylis candicans* (Smith et al., 2011), (d) *Lotus japonicus* wild type (Burak et al., 2021), (e) *Stipagrostis ciliata* (Marasco et al., 2022b), (f) *Xyris* sp. (Oliveira et al., 2015), (g) Mutant genotypes without root hairs of *Lotus japonicus* (Burak et al., 2021), (h) *Heteropappus altaicus* (photographed by the authors in October 2022). These images are reproduced with permission from Springer Nature, Oxford University Press, and John Wiley and Sons. We also recommend Abrahão et al. (2014), Hanna et al. (2013), Delhaize et al. (2012), Shane et al. (2010) for good photo illustrations of rhizosheaths.

size in drought environments but would show negligible rhizosheaths growing in water-rich conditions (Zhang et al., 2020).

As a special root–soil interface trait, rhizosheath has attracted interdisciplinary attention. Pang et al. (2017), Ndour et al. (2020), and Aslam et al. (2022) have addressed multiple factors underlying rhizosheath formation and the significance of rhizosheaths in water and nutrient acquisition from different perspectives. These important reviews have introduced what rhizosheaths are, why plants form rhizosheaths, and what rhizosheaths can do, particularly in the field of future sustainable agriculture. However, the terminology, methodology, formation mechanisms, and findings of the features and functions of rhizosheaths relative to other soils have not yet reached a consensus, which hinders our progress in understanding and harnessing the environmental benefits of rhizosheaths. Here, we conducted a comprehensive literature survey and explored the updated understanding of rhizosheaths in recent years via an integrated and quantitative approach. By addressing the terminology, methodology, mechanisms underlying rhizosheath formation, and findings of the features and functions of rhizosheaths, this systematic review aims to improve and advance the understanding of rhizosheaths by integrating the results of interdisciplinary studies and discuss the multiple roles of rhizosheaths in environmental processes from systematic and quantitative perspectives.

2. Methodologies and terminologies about rhizosheaths

Upon a comprehensive literature survey of relevant publications during the last 5 years, we summarize the definitions of rhizosheaths and the methods used to collect rhizosheath samples (Table 1). Rhizosheath is generally defined based on the adhesiveness of soil to root systems; however, the collection methods of rhizosheath soil vary in different studies. Some studies used vigorous shaking, brushing, and hand-kneading to collect rhizosheath soils (Kong et al., 2018; Rabbi et al., 2018a; Liu et al., 2021; Hallama et al., 2022; Mo et al., 2022a). However, due to the strong adhesion between the rhizosheaths and root systems, it is not easy to collect the intact rhizosheaths without destroying the root system by such methods. It is also challenging to maintain the consistency of such methods across different studies because this manual procedure is usually described ambiguously. Even if the same method is applied, the strength of shaking, brushing, or hand-kneading performed by different researchers is inconsistent, resulting in questionable comparability and reliability among research results. Noted, the adhesiveness of rhizosheath soils to the roots also varies across species (i.e., different strength scores as proposed by Brown et al. 2017), suggesting that the collection of rhizosheath soils can be more accessible for some species than others; therefore, the same “shaking strength” may not fit in every case. Considering such problems, some other studies used an improved method, i.e., collecting rhizosheath by washing the roots with purified water and then recovering it by oven drying (Rabbi et al., 2022; Ndour et al., 2022; Burak et al., 2021; Zhang et al., 2020; Basirat et al., 2019). Such improved collection methods allow for collecting most of the rhizosheath soils with relatively higher consistency and not breaking root systems. It is also considered necessary for the complete and accurate collection of rhizosheath soils (Rabbi et al., 2022). The commonly used methods of rhizosheath collection outlined in Table 1 provide a comprehensive summary for future studies to refer to and may help researchers choose suitable collecting methods up to specific scientific questions. Nevertheless, precise and feasible sampling of rhizosphere soils remains a great challenge as each method has pros and cons. Future work would be required to resolve this difficulty.

Plant root–soil–microbe interactions have long been a research hotspot due to the invisibility and complexity of the belowground ecosystem. Despite numerous studies in root and belowground ecology, the presence of rhizosheaths has not been widely reported. One of the important reasons is that researchers are likely to regard rhizosheath soils as rhizosphere counterparts. The rhizosphere is conceptualized as

Table 1

Definitions and collection methods of rhizosheath.

Definition of rhizosheaths	Collection Method	Pros	Cons	References
Soil layer attached to root surface	The collected roots with rhizosheath were immersed in purified water and shaken overnight at 100 rpm, and then dried to constant weight at 40 °C	(1) Feasible and repeatable; (2) Collecting most of the rhizosheath soils; (3) Not breaking root systems.	(1) Indirect collection with potential loss of water-extractable fractions; (2) Not suitable for microbial analyses.	Rabbi et al., 2022
Root-adhering soil	Rhizosheath soil was collected by washing the roots using distilled water, and then was dried to constant weight at 40–105 °C			Ndour et al., 2022; Burak et al., 2021; Zhang et al., 2020; Basirat et al., 2019
Sheath-like soil that tightly coats the root surface	Rhizosheath soil was physically separated from the roots by vigorous shaking and carefully hand-kneading	(1) Feasible; (2) Direct collection without potential loss of water-extractable fractions.	(1) Breaking root systems; (2) Intact rhizosheath soils are hard to collect; (3) Poor repeatability due to varied shaking strength;	Mo et al., 2022a; Mo et al., 2022b; Kong et al., 2018
Root-adhering soil	Rhizosheath soil was physically separated from the roots by gently brushing off the soil from the root surface, and then dried to constant weight at 40 °C		(4) Not suitable for microbial analyses.	Hallama et al., 2022; Liu et al., 2021; Rabbi et al., 2018a; Rabbi et al., 2018b; Rabbi et al., 2021
Root-adhering soil which encases the entire root system	Rhizosheath soil was excised and physically separated from the inner root tissues using a sterile scalpel	(1) Suitable for microbial analyses; (2) Direct collection.	(1) Breaking root systems; (2) Collect only a portion of rhizosheath soils.	Marasco et al., 2018; Marasco et al., 2022b

the soil zone affected by roots via rhizodeposits, microbes, and other pathways (Hiltner, 1904). Rhizosphere soil includes soil attached to the root surface to varying degrees (Freschet et al., 2021). As either the size or the shape of the rhizosphere is challenging to quantify, researchers usually take root-adhering soil as rhizosphere soil for field studies because root-adhering soil has relatively clear boundaries from the surrounding environments and is relatively easy to sample (Hallett et al., 2022). Although root-adhering soil is a part of rather than the whole rhizosphere, it is a typical representative of rhizosphere soil due to its close contact with root systems (Pang et al., 2017). We revisit previous studies on root-adhering soil and found that “root-adhering soil” mentioned in the literature, or “rhizosphere soil” collected using the methods in Table 1, is mostly rhizosheath soil (e.g., Czarnes et al.,

2000a; Bogeat-Triboulot et al., 2004; Sun et al., 2018). As such, there is a growing consensus that “root-adhering soil” is synonymous with “rhizosheath soil” for plants with rhizosheaths (Rabbi et al., 2018a; Ndour et al., 2020; Ndour et al., 2021).

Despite these advances, it is essential to distinguish “rhizosheath soil” from other similar terms to clarify the conceptual confusion. Based on our understanding of the current terminologies of the rhizosheath and relevant concepts, we propose a terminology framework as illustrated in Fig. 2. If plants do not form rhizosheath, which suggests that there is negligible soil attached to the root surface upon excavation and hand-shaking, the soil falling off the root surface after gentle shaking should be designated as “loose rhizosphere soil” or “non-rhizosheath rhizosphere soil”. In this case, the collection method of loose rhizosphere soil of NRF plants should refer to the common procedures used for sampling rhizosphere soil (shaking roots or soaking roots in liquid for soil detachment) depending on the specific research question (Barillot et al., 2013; Freschet et al., 2021). In cases where plants form rhizosheath—suggesting that root-adhering soil can be clearly observed after excavation and hand-shaking, the soil that has been affected by roots and falls off the root surface after gentle shaking should be designated as “outer rhizosphere soil”, while the remaining soil that is still firmly attached to the root surface should be designated as “inner rhizosphere soil” (i.e., “rhizosheath soil” or “root-adhering soil”). The difference between NRF and RF situations is that the former has negligible adhering soil particles after gentle shaking. Of note, the loose rhizosphere soil of NRF plants (hereafter “loose rhizosphere soil”) and the outer rhizosphere soil of RF plants (hereafter “outer rhizosphere soil”) are mostly the same in the collection method because both are loosely attached to the root systems. However, they are specifying different kinds of situations (NRF vs RF) and are different in the ranges.

Loose rhizosphere soil of NRF plants refers to the whole rhizosphere because the rhizosheath is absent, while the outer rhizosphere soil of RF plants refers to the part of rhizosphere outside rhizosheaths. Whether the characteristics of loose and outer rhizosphere soils differ from each other may depend on the plant species and the distance of sampling soils from roots.

3. Mechanisms underlying rhizosheath formation

Mechanisms and influencing factors underlying rhizosheath formation have long been a research focus of rhizosheath. Current studies have shown that rhizosheath formation depends primarily on root traits (e.g., root hair and root mucilage), microbial activity associated with plant roots (e.g., microbial-derived mucilage and mycorrhizal association), and soil conditions and physio-chemical processes (e.g., soil texture, soil moisture, and dry-wet cycles). Among them, root hair traits (e.g., amount, length, and density) and root- and microbial-derived mucilage are the key drivers regulating rhizosheath formation.

First, root hairs provide important conditions for rhizosheath formation and an essential basis for its environmental functions (Opoku et al., 2022). Root hairs are tubular subcellular protrusions of the root epidermal cells (Dolan & Costa, 2001) and are the main sites for water and nutrient uptake (Parker et al., 2000). The presence of root hairs significantly increases the contact area of root-soil interfaces, allowing more soil particles to attach to the root surfaces and hence providing a physical framework for rhizosheath formation (Haling et al., 2010). Theoretically, longer root hairs are likely to be associated with larger rhizosheaths, but results from current studies are not sufficient to confirm such a correlation. Some studies have found that the rhizosheath size (relative weight, dry weight, and specific weight) of

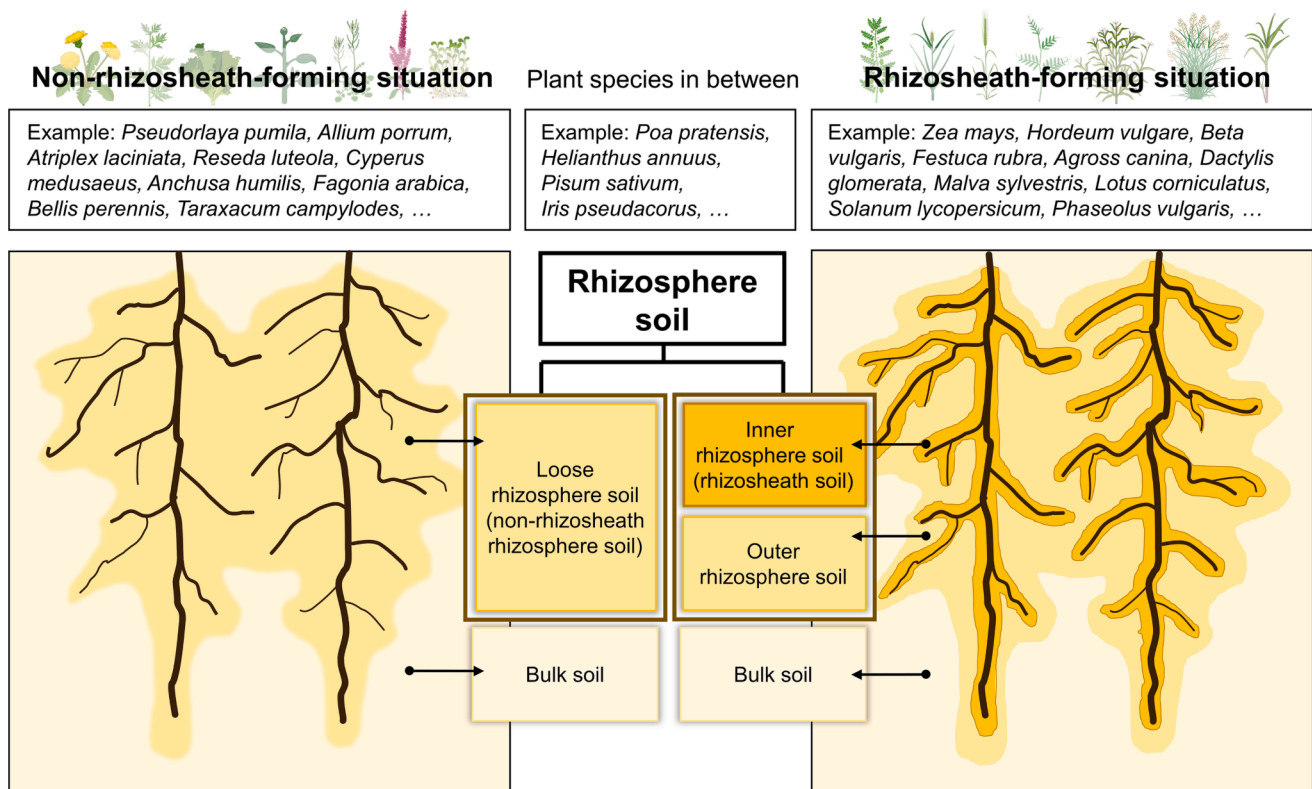


Fig. 2. Proposed terminology framework for the definitions of loose rhizosphere soil (non-rhizosheath rhizosphere soil) of non-rhizosheath-forming (NRF) plants, inner rhizosphere soil (rhizosheath soil) of rhizosheath-forming (RF) plants, outer rhizosphere soil of RF plants, and bulk soil. We propose that for NRF situation, all rhizosphere soils that are loosely attached to the roots are termed as “loose rhizosphere soil” or “non-rhizosheath rhizosphere soil”; for RF situation, rhizosphere soil consists of two parts: the soil that is tightly attached to the roots is termed as inner rhizosphere or rhizosheath soil; the soil that is loosely attached to the roots is termed as outer rhizosphere soil. Examples of plant species that generally do not form rhizosheaths, clearly form rhizosheaths, and occasionally form rhizosheaths are listed according to Brown et al. (2017).

wheat, cowpea, and other species have shown a strong positive correlation with the length and density of root hairs (Delhaize et al., 2015; Opoku et al., 2022). However, a considerable number of studies have reported a weak correlation between the rhizosheath size of some plant species (e.g., barley and chickpeas) and their root hair lengths (Haling et al., 2010; George et al., 2014). Further, whether the presence of root hairs is necessary for rhizosheath formation remains arguable because limited size rather than a complete absence of rhizosheaths is observed in root hairless mutants (George et al., 2014; Burak et al., 2021) and rhizosheaths are not necessarily observed where root hairs are present (Fig. 3c, *Potentilla acaulis*).

More quantitative impacts of root hair length on rhizosheath formation have been recently proposed. For example, in the presence of root hairs with a length of <300 µm, root hair length is the primary factor controlling rhizosheath size and stability; additionally, significant root hair development contributes to rhizosheath formation (Rongsawat et al., 2021). However, in the absence of root hairs or when the root hair length exceeds 300 µm, mucilage and other root traits (e.g., root hair density and morphology) become the dominant factors (Burak et al., 2021; Rongsawat et al., 2021). Further field experiments using broader genotypes might help verify the postulated correlations. Including more crop varieties in studies might also be helpful to provide meaningful data for future crop breeding and solid evidence for the impacts of root hairs (Hallett et al., 2022). By integrating current research results, we suppose that root hair trait cannot fully explain the variation in rhizosheath production, and the correlation between the length of root hairs and rhizosheath size may be affected by the amount and density of

root hairs, as well as the complementary effects of other influencing factors controlling rhizosheath formation.

Compared to the physical adhesion of soil particles by root hairs, mucilage exhibits a stronger binding capability to soil particles. Therefore, the relative contribution of mucilage can sometimes predominate rhizosheath formation. There are two primary sources of mucilage in rhizosheaths: root mucilage (root exudates) and microbial-derived mucilage (Marasco et al., 2018). Root mucilage is mainly secreted by root caps and root epidermis (Carminati and Vetterlein, 2013), while microbial-derived mucilage is mainly produced during microbial activities (Czarnes et al., 2000b). Both root- and microbial-derived mucilage are mainly composed of polysaccharides, proteins, and lipids (Nazari, 2021). This hydroxyl-rich mucilage can bind with soil particles through hydrogen bonding, which primarily contributes to soil aggregation around roots (Albalasmeh and Ghezzehei, 2014). Moreover, mucilage significantly influences soil water dynamics and dry-wet cycles in the rhizosphere microenvironment. The drying of mucilage is considered a key driver of soil adhesion (Watt et al., 1994). During soil wetting, soil pores are filled with water; during soil drying, the mucilage transforms into a high-viscosity adhesive “gel” due to water loss and decreased surface tension, thereby connecting soil particles more tightly around the root systems (Aslam et al., 2022). Long-term drought conditions can trigger the release of mucilage in larger quantities (Albalasmeh and Ghezzehei, 2014), enhance the adhesiveness of mucilage and root hair formation through intensified dry-wet cycles, thereby creating a more stable rhizosheath structure (Aslam et al., 2022; Carminati et al., 2010).

Scanning electron microscopy images have confirmed that soil

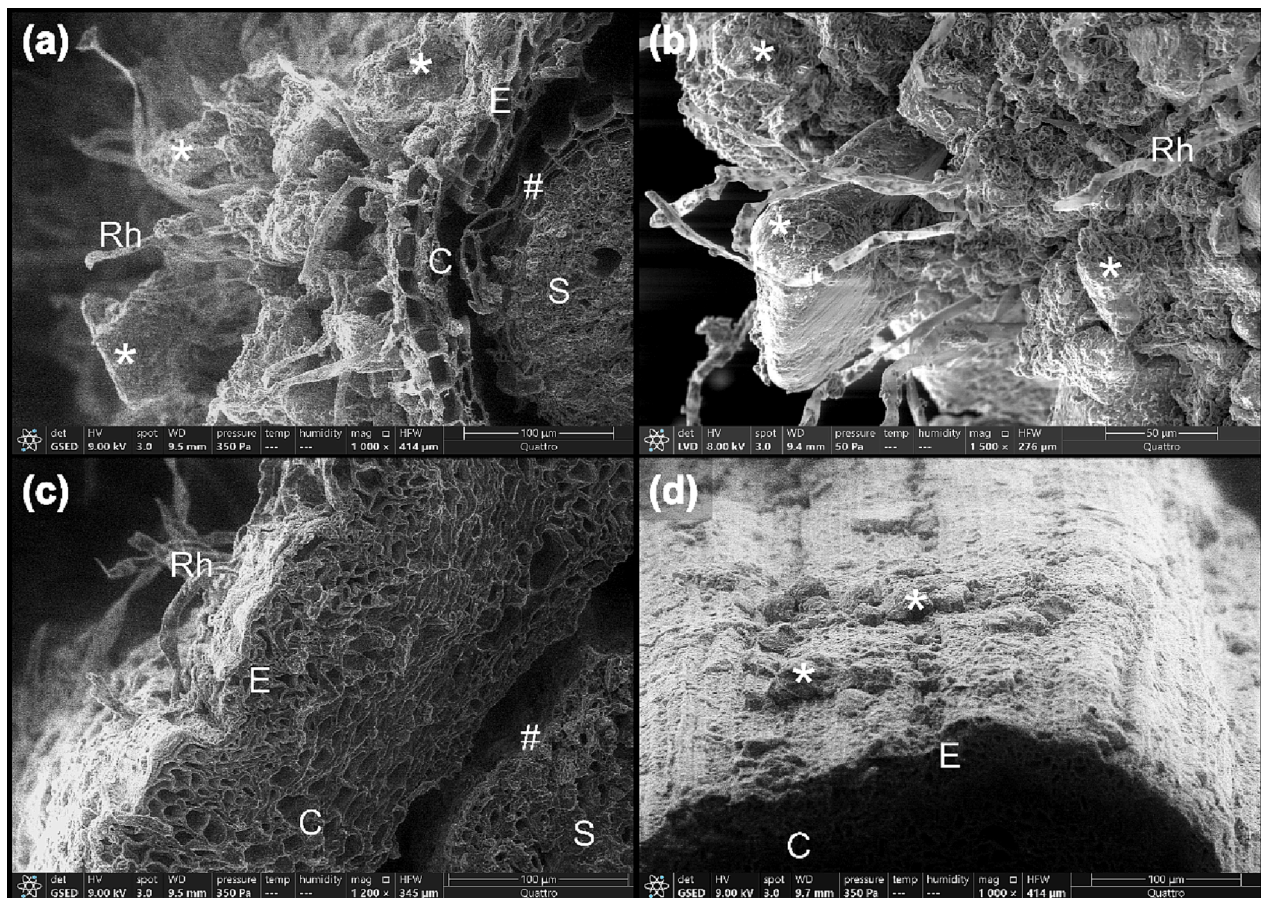


Fig. 3. Visualization of different external root layers with and without rhizosheaths using environmental scanning electron microscopy (ESEM). (a) External root layer of typical rhizosheath-forming plant, *Cleistogenes squarrosa*. (b) External root layer of typical rhizosheath-forming plant, *Stipagrostis pennata*. (c) External root layer of typical non-rhizosheath-forming plant, *Potentilla acaulis*. (d) External root layer of typical non-rhizosheath-forming plant, *Heteropappus altaicus*. (C) Cortical tissue. (E) Epidermis. (Rh) Root hairs. (S) Stele. (*) Sand grain surface. (#) Endodermis: layer of cells between stele and cortical tissues. Methods of visualizing external root layers using ESEM were presented in Text S2.

particles in the rhizosheath are tightly wound by root hairs (Fig. 3a-b, *Cleistogenes squarrosa* and *Stipagrostis pennata*), with apparent mucilage on the surface of root hairs (Marasco et al., 2018). The more organic matter (especially polysaccharides) contained in mucilage, the thicker rhizosheaths are likely to be (Albalasmeh & Ghezzehei, 2014). Specific compounds, such as xyloglucan, heteroxylan, glycoprotein epitopes, and some low-molecular-weight metabolites, are also likely to contribute to the soil-binding capabilities of mucilage, thereby affecting rhizosheath formation and development (Galloway et al., 2022). As the composition and quantity of root and microbial-derived mucilage vary significantly with species and environmental conditions, the soil-binding capacities of plant roots to soil particles may also exhibit great variations (Galloway et al., 2022), resulting in high heterogeneity of rhizosheath structure across species. Further studies are needed to elucidate the contribution of different molecular components of mucilage to the formation and development of rhizosheaths.

Rhizosheath formation is a complex process associated with many factors aside from root hairs and mucilage. On the one hand, mycorrhizal fungi can influence rhizosphere stability through root extension by hyphae and the release of glycoproteins (Peng et al., 2013). It also contributes to rhizosheath formation together with root hairs and mucilage; however, the contribution from mycorrhizal associations was much minor compared to root hairs and mucilage as noted in previous studies (Moreno-Espíndola et al., 2007; Pang et al., 2017; Ndour et al., 2020). No correlation between rhizosheath thickness and arbuscular mycorrhizal fungal colonization level was found across different grass species (Hartnett et al., 2013). While hyphae can help with soil aggregation, mycorrhizal fungi can also potentially reduce root exudation and root hair length (Ryan et al., 2016, 2012). Therefore, mycorrhizal fungi are likely to be involved in rhizosheath formation through quite complex processes.

On the other hand, soil processes (e.g., dry-wet cycles) and associated abiotic factors (e.g., water content and texture) also have significant influences on rhizosheath formation. Dry-wet cycles interact with gluing agents in the rhizosphere (i.e., mucilage, as previously mentioned), playing complementary roles in promoting rhizosheath formation (Aslam et al., 2022). Such processes are also influenced by soil water content. Rhizosheath formation can occur in different manners at two extremes of the soil water content axis, i.e., larger rhizosheath size in drought environments versus smaller rhizosheath size or even negligible rhizosheaths in water-rich environments (Watt et al., 1993; Zhang et al., 2020). Moreover, compared to loamy and clayey soils, sandy soils—characterized by lower porosity and poorer root-soil contact—tend to create larger rhizosheaths (Hallett et al., 2022). Taken together, drought stress and sandy soils seem to favor rhizosheath formation. This might be one of the possible reasons that rhizosheaths are prevalent in arid and semiarid ecosystems, especially in xerophytes (Bailey & Scholes, 1997; Iijima et al., 2004). Studies have evidenced that plants induced by drought stress have more rhizosheath cover (Watt et al., 1994; Rabbi et al., 2021). Basirat et al. (2019) also showed that under dry conditions, plant cultivars with greater rhizosheaths sustained a higher transpiration rate, while those with lower rhizosheath formation suffered from drought stress and reached an earlier wilting point. Therefore, rhizosheaths are considered to be the adaptive root-soil interface traits developed by plants to buffer themselves against abiotic stress such as long-term drought and desertification (Hallett et al., 2022).

While it is evident that root hairs and mucilage contribute significantly to rhizosheath formation and development, it is worth noting that root hair traits and mucilage are associated because the presence of root hairs could promote the release of root mucilage, especially soil-binding polysaccharide complex (Galloway et al., 2022). Burak et al. (2021) found that there were complementary and additive effects of root hairs and root mucilage on rhizosheath formation. As such, further work should pay more attention to the inter-associations between root hair traits, mucilage chemistry, microbial profiles, and soil environmental

characteristics. There are also increasing studies trying to unravel the genetics underlying rhizosheath formation. A series of genes and quantitative trait loci associated with rhizosheath formation have been identified, which may allow for harnessing the drought tolerance capacity of rhizosheaths via these transcription factors in future agriculture (Aslam et al., 2022). Although it is acknowledged that rhizosheath is a combination of genetic and environmental factors traits (Delhaize et al., 2015; George et al., 2014), it remains controversial which one is the dominant factor, i.e., whether rhizosheath is more plant species-specific (genetic-controlled) or more environment-controlled. Further exploration of the multiple effects of physical, chemical, biological, and genetic factors that regulate rhizosheath formation and development is required.

4. Quantification of the differences between rhizosheath soils and other soils

4.1. Physical characteristics of rhizosheath soils

One of the most important characteristics of rhizosheath soils is their water content, which has become a research interest since 1995; early results have shown a significantly higher water content in rhizosheath soils than in bulk soils (Young, 1995). This observation still holds when comparing rhizosheath (inner rhizosphere soil) with outer rhizosphere soils and bulk soils (North and Nobel, 1997; Moreno-Espíndola et al., 2018). Increasing publications supported this distinctive characteristic of rhizosheath soils; therefore, we attempted to quantify exactly how much the water content in rhizosheath soils is higher than in other soils. To this end, we collected evidence reported in the literature and performed a statistical analysis, which provided a quantitative overview of the differences between rhizosheath soils and other soils (Fig. 4), including outer rhizosphere soils, loose rhizosphere soils, and bulk soils as defined in Fig. 2. The source data are presented in Table S1, and the methods of data collection and processing are presented in Text S1. Statistical results showed that, on average, the gravimetric water content in rhizosheath soils was 66% higher than that in outer rhizosphere soils and bulk reference soils, based on 40 paired records from 14 studies ($p < 0.01$; Fig. 4). Specifically, the average gravimetric water content was 16% in rhizosheath soils according to these studies, while it was 11%, 10%, and 9% in non-rhizosheath rhizosphere soils (loose rhizosphere soils), outer rhizosphere soils, and bulk soils, respectively.

Porosity is another important physical characteristic of soil. Some studies suggested that new root proliferation and aerial tissue development might create more pores in the root-soil interface (Helliwell et al., 2019; Zhang et al., 2020). Statistical results showed that, on average, porosity in rhizosheath soils was 11% higher than that in bulk soils ($n = 16$; $p < 0.01$; Fig. 4). Such results allied with some previous observations, which suggested that rhizosheath was more porous with more small soil particles and more connected macro-pores than the surrounding bulk soil (Ma et al., 2011; Rabbi et al., 2021; Rabbi et al., 2018a). Noted, it was suggested that drought-tolerant species are more likely to alter soil porosity compared to drought-sensitive species (Rabbi et al., 2018a). A decrease in porosity closer to the root surface could sometimes occur due to variations in plant species or experimental settings (Young, 1998; Bruand et al., 1996).

4.2. Chemical characteristics of rhizosheath soils

The strong root-soil-microbe interactions can potentially modify the chemical characteristics of rhizosheath soils. Root exudates can directly affect rhizosheath formation and size (Ndour et al., 2021). Particularly for legumes with rhizosheaths, the release of organic acids via root exudation can stimulate the activity of nitrogen-fixing microorganisms and reduce the uptake of inorganic nitrogen by the roots (Tian et al., 2021). As organic acids are usually exuded in the anionic form (e.g., carboxylate), the great root exudation does not necessarily cause

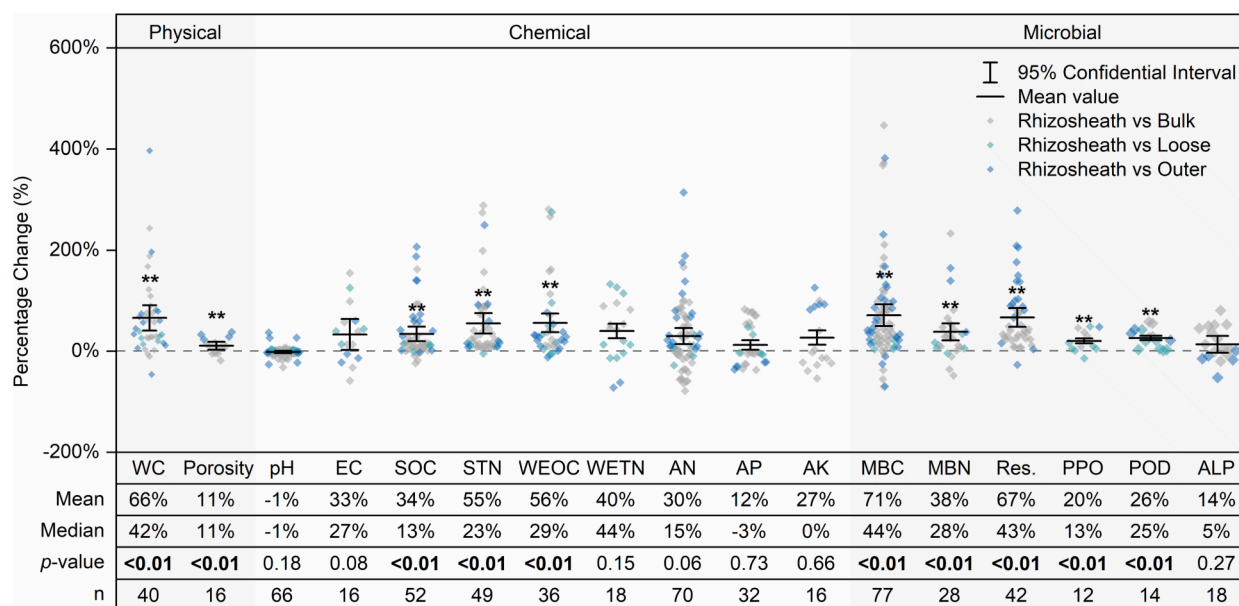


Fig. 4. Different physical, chemical, and microbial features of rhizosphere soils from bulk soils (grey dots), loose rhizosphere soils (green dots), or outer rhizosphere soils (blue dots) as indicated by the percentage changes. The percentage changes are calculated as: parameter of rhizosphere soils – parameter of reference soils/parameter of reference soils $\times 100\%$. In Fig. 4, the paired soil as reference in the literature included outer rhizosphere soil of RF plants, loose rhizosphere soil of NRF plants, or bulk soil (Table S1). The abbreviations in Fig. 4 refer to: WC, gravimetric water content; EC, electrical conductivity; SOC, soil organic carbon content; STN, soil total nitrogen content; WEOC, water-extractable organic carbon content; WETN, water-extractable total nitrogen content; AN, available nitrogen content; AP, available phosphorous content; AK, available potassium content; MBC, microbial biomass carbon content; MBN, microbial biomass nitrogen content; Res., respiration rate; PPO, polyphenol oxidase activity; POD, peroxidase activity; ALP, alkaline phosphatase activity. Error bars represent 95% confidential intervals (CIs) of percentage changes. When 95% CIs do not overlap zero and the p -values from random effect models are <0.05 , the effects are considered significant (*, $p < 0.05$, statistically significant; **, $p < 0.01$, highly significant). Methods of data collection and processing are presented in Text S1. Source data of these variables and their references are presented in Table S1.

significant changes in rhizosphere soil pH (Hinsinger et al., 2003), which coincides with our statistical results. Our results showed that there was no significant difference in pH values between rhizosphere and other soils (mean percentage change = -1% ; $n = 66$; $p = 0.18$; Fig. 4). In some cases, the pH of rhizosphere soils was even higher than that of outer rhizosphere soils or bulk soils (e.g., Mo et al., 2022a; Prendergast-Miller et al., 2014). As an important indicator of soil salinity, electrical conductivity (EC) did not significantly differ between rhizosphere soils and other soils (mean percentage change = $+33\%$; $n = 16$; $p = 0.08$). Although some studies proposed that rhizosphere soils contain a relatively lower water-soluble salt content, providing a relatively low osmotic pressure environment for plant roots, which was conducive to plant water acquisition from the rhizosphere (Ren et al., 2018; Luo et al., 2013), current results were not sufficient to confirm such speculations.

Moreover, compared with rhizosphere soils and bulk soils, rhizosphere soils contain higher soil organic carbon (including total organic carbon, dissolved organic carbon, and microbial biomass carbon) and nitrogen (including total nitrogen, dissolved organic nitrogen, and microbial biomass nitrogen) contents (Fig. 4). Statistical results showed that, on average, the soil organic carbon (SOC) content in rhizosphere soils was 34% higher than that in other soils based on 52 paired records from 18 studies ($p < 0.01$), and the soil total nitrogen (STN) content in rhizosphere soils was 78% higher than that in other soils based on 49 paired records from 13 studies ($p < 0.01$). For the water-extractable fractions, the water-extractable organic carbon (WEOC) content in rhizosphere soils was, on average, 56% higher than that in loose rhizosphere soils or bulk soils ($n = 36$; $p < 0.01$), and the water-extractable total nitrogen (WETN) content in rhizosphere soils was, on average, 40% higher than that in loose rhizosphere soils or bulk soils ($n = 18$; $p = 0.15$). These results reveal a significantly higher carbon content in rhizosphere soils than in other soils, and such differences were particularly evident in the water-extractable carbon fractions. Our previous study also found that compared to loose rhizosphere soils,

rhizosphere soils have significantly greater carbon input from plant and microbial sources (Mo et al., 2022a), as well as a higher content of labile components (e.g., water-extractable organic matter) and greater bioavailability (Mo et al., 2022b). An increasing number of studies suggest that RF roots release more carbon into the soil, and rhizosphere soils provide a more favorable environment for soil organic matter (SOM) accumulation (Moreno-Espíndola et al., 2018; Pang et al., 2022). Specifically, the contents of free lipids, hydrolyzable lipids, and lignin phenols were 16%, 24%, and 22% higher in rhizosphere soils than in loose rhizosphere soils, respectively (Mo et al., 2022a). The relative abundances of humus-like and protein-like fluorescent dissolved organic matter were 55% and 152% higher in rhizosphere soils than in loose rhizosphere soils, respectively (Mo et al., 2022b). It is worth noticing that Zhao et al. (2022) conducted a meta-analysis that showed that the WEOC content in rhizosphere soils was 5% higher than those in bulk soils in the grassland. The difference in WEOC contents between rhizosphere soils and bulk soils is far larger than that between generalized rhizosphere soils and bulk soils, as reported by Zhao et al. (2022), which further supported the distinct labile organic matter characteristics of rhizosphere soils and the necessity to consider different rhizosphere compartments.

In addition to organic substances, some studies have reported differences in the inorganic nutrients between rhizosphere and other soils. Statistical results showed that, on average, the contents of available nitrogen (AN), available phosphorous (AP), and available potassium (AK) were 30% ($n = 70$; $p = 0.06$), 12% ($n = 32$; $p = 0.73$), and 27% ($n = 16$; $p = 0.66$) higher in rhizosphere soils than in other soils, respectively. Although the differences in these mineral nutrient contents between rhizosphere and other soils were not statistically significant, their 95% CIs did not overlap zero and there have been a considerable number of studies suggesting that increased nutrient uptake efficiency is associated with rhizosphere formation and development (Haling et al., 2016; Aslam et al., 2022). Hallama et al. (2022) and Pang et al. (2022) both

reported significantly higher content of microbial biomass phosphorus in rhizosheath soils than in outer rhizosphere soils or bulk soils. Kong et al. (2018) found that the content of total phosphorus in rhizosheath soils was ~55% higher than that in bulk soils. These differences in chemical characteristics between rhizosheath and other soils are the essential basis for higher water and nutrient uptake and retention in rhizosheath soils than in other soils.

4.3. Microbial characteristics of rhizosheath soils

Rhizosheaths are widely distributed in arid and semiarid ecosystems, where prolonged drought stress can lead to a decreased abundance and activity of rhizosphere microbial communities (Sanaullah et al., 2011). Adaptive evolution of the rhizosheaths has led to the recruitment and selection of unique microbial taxa in rhizosheath soils, including some plant growth-promoting rhizobacteria that differ significantly in the microbial abundance, species, and activity from those in other soils (Othman et al., 2004; Marasco et al., 2022a). Despite the limited number of observations, we noticed that microbial biomass and soil respiration rates were significantly higher in rhizosheath soils than in outer rhizosphere soils or bulk soils (Fig. 4). The microbial biomass carbon (MBC) and microbial biomass nitrogen (MBN) in rhizosheath soils were, on average, 71% ($n = 77$; $p < 0.01$) and 38% ($n = 28$; $p < 0.01$) higher than that in other soils, respectively; the soil respiration rate in rhizosheath soils was, on average, 67% higher than that in bulk soils ($n = 42$; $p < 0.01$; Fig. 4). Several studies have reported a considerably higher abundance of specific microbial communities in rhizosheath soils than those in other soils (Marasco et al., 2018; Ndour et al., 2020), including bacteria, fungi, and actinomycetes, with the most significant differences noticed in bacterial communities (Hallama et al., 2022). Those microbial communities were also found to be associated with the rhizosheath size. Ndour et al. (2021) reported significant correlations between the characteristics of soil microbial communities (e.g., microbial biomass, bacterial β -diversity, soil enzyme activity, and specific richness) and the rhizosheath size. The enrichment of specific fungal communities, such as *Trichoderma* spp. and *Hebeloma cylindrosporum*, was also considered to contribute to larger rhizosheath size by increasing the amount of the polysaccharide in the rhizosphere and other potential mechanisms (Bochicchio et al., 2022; Ndour et al., 2021; Bogeat-Triboulot et al., 2004). Nevertheless, the causal effects between the microbial communities and rhizosheath features remain an important issue to be verified.

Soil enzymes are essential for organic matter transformation and plant nutrient uptake. The production and distribution of soil enzymes in the rhizosphere directly affect the utilization of nutrients by plants (Kuzakov, 2002). Statistical results showed that soil enzymatic activities were significantly higher in rhizosheath soils than in other soils (Fig. 4). For example, the activity of polyphenol oxidase (PPO), which is mainly derived from soil microorganisms, root exudation, and the decomposition of plant residues (Mayer, 2006), was, on average, 20% ($n = 12$; $p < 0.01$) higher in rhizosheath soils than that in other soils (Fig. 4). Similarly, the activity of peroxidase (POD), which can be produced by fungi and is associated with plant stress resistance and degradation of polymers (Datta et al., 2017; Sinsabaugh, 2010), was, on average, 26% ($n = 14$; $p < 0.01$) higher in rhizosheath soils than that in other soils (Fig. 4). In addition, on average, the activity of alkaline phosphatase (ALP) in rhizosheath soils did not significantly differ from that of other soils (mean percentage change = +14%; $n = 18$; $p = 0.27$). Noted, while PPO and POD are oxidoreductases catalyzing redox reactions, ALP is an extracellular hydrolase which catalyzes hydrolysis reactions (Mayer, 2006). PPO and POD are suggested to be less stable in soil than extracellular hydrolases, especially when associated with organic particles (Sinsabaugh, 2010). This may explain the smaller variations of ALP than that of PPO and POD between rhizosheath soils and other soils.

5. Environmental functions of rhizosheaths

5.1. Improvement of soil architecture

Plants have long been considered the important architects of soil (Hallett et al., 2022). Plants with rhizosheaths are thought to be well adapted to extreme environments, which are frequently characterized by long-term drought, soil and wind erosion, and desertification (Othman et al., 2004; Danin, 1996). Recent evidence further validates the roles of rhizosheaths in maintaining and improving soil architecture. First, the formation of rhizosheath forms a clear boundary that separates inner rhizosheath soil from outer rhizosphere soil. The sheath-like structure creates a buffer barrier between plant roots and the surrounding environment, which can protect plant roots from suffering surface heat stress and friction damage caused by drifting sands (Wang et al., 2022). The growth of RF plants in mobile and semi-mobile sand dunes can also gather the mobile sand particles around roots and enhance root anchoring to the loose sandy substrate (Liu et al., 2019). Therefore, the formation and development of rhizosheaths render the collapse of RF plants under wind-sand erosion less likely, enabling RF plants to adapt to these stressful environments and inhabit as pioneer species (Tian et al., 2019). Typical RF plants, including *Stipagrostis pennata*, *Kengyilia hirsuta*, *Leymus secalinus*, and *Elymus nutans*, show great potential in sand fixation and are commonly the pioneer grasses in arid and semiarid ecosystems due to their adaptation to environmental stresses (Wang et al., 2022).

Second, although rhizosheath does not seem to alter soil porosity or soil texture remarkably, it is widely acknowledged that soil aggregation around RF roots is enhanced because rhizosheath formation is the direct cause of soil aggregation around roots due to root hairs, adhesive mucilage, and soil environment (Moreno-Espindola et al., 2007; Marasco et al., 2018). Previous studies have noticed that rhizosheath soils exhibit a greater tensile strength of soil aggregates than bulk soils (Czarnes et al., 2000b). In arid and semiarid ecosystems characterized by severe soil erosion, poor soil stability, and weak inter-bonding of soil particles, rhizosheaths have great potential in promoting soil aggregation and stabilizing soil structure. As these functions are environmentally critical for building a healthy soil architecture, the application of RF plants in ecological restoration should be well considered and evaluated.

5.2. Improvement of resource uptake and availability

A consensus has been reached that rhizosheaths can improve resource (i.e., water and nutrients) uptake and availability by roots, thus assisting plant growth. Therefore, rhizosheath is considered a strategy for plants to cope with resource-poor environments and is known as a potential "resource reservoir" for plant roots. The detailed mechanisms by which rhizosheaths enhance resource uptake and availability remain to be discussed. Under water deficit, roots tend to shrink radially, which creates root-soil air gaps and results in poor root-soil contact (North and Nobel, 1997). For RF plants, the formation of the sheath-like structure physically mitigates the development of air gaps around roots, and thus protecting root-soil contact and the potential for water uptake (Pang et al., 2017).

Moreover, the key factors regulating rhizosheath formation, root hairs and mucilage, also contribute to the functions of rhizosheath in resource uptake and availability. On the one hand, the development of root hairs can penetrate the soil and help create a more porous soil layer at the root-soil interface, which can enhance the water-holding capacity in rhizosheath soils (Ma et al., 2011). Longer root hairs can also penetrate the soil and greatly enhance the accessibility of roots to soil pore water (Wissuwa & Kant, 2021), which helps maintain soil particle connectivity. This would break the continuous water film in the soil pores, and thus reducing evaporative water loss from soil under drought conditions (Rabbi et al., 2021). On the other hand, the highly viscous

mucilage can serve as a surfactant to reduce water surface tension by more than 30% (Read et al., 2003), which promotes water uptake by roots from the surrounding soil. It can also serve as a hydrogel to resist water drainage and improve water retention in rhizosheath soils (Zhang et al., 2021).

While root hairs and mucilage interactively aid efficient water uptake and retention, nutrient acquisition and availability also benefit from the microhydrological niches created by rhizosheaths (Benard et al., 2019), because nutrient acquisition is highly dependent on the mass flow of water to the root surface (Chapman et al., 2012) and mineral nutrients usually dissolve in water and then can be absorbed by roots (Lambers & Oliveira, 2019). Moreover, rhizosheaths were suggested to associate with nitrogen fixation. Over 40% of identified soil microbes were associated with nitrogen fixation (Tian et al., 2019). There was also evidence showing that rhizosheath soils have higher linear acetylene-reducing activity than bulk soils (Hanna et al., 2013) and can accumulate a large number of the nitrogen-fixing microbiome (Tian et al., 2019), both of which help plants to better withstand nutrient deficits. Altogether, rhizosheaths provide a favorable living environment for roots and microorganisms by supplying essential water and nutrients, thereby creating a “mini-oasis” (Marasco et al., 2022b). An efficient water and nutrient uptake may suggest a good performance of plant growth and health. This may further enhance plant defense against soil pathogens and contaminants (Bochicchio et al., 2022; Wang et al., 2022). The selection of key traits associated with these functions may aid in the artificial modification of plant stress resistance and improve crop yields at the lowest metabolic cost (Burak et al., 2021; Haling et al., 2016).

5.3. Acceleration of soil organic carbon turnover

SOM derived from plants and microbes significantly affects the soil environment (Lehmann & Kleber, 2015). The formation of rhizosheaths represents a considerable resource investment into belowground by plants due to the extended root hairs and greater exudation of mucilage (Holz et al., 2018; Ndour et al., 2020). That is, substantial amounts of photosynthesized carbon are allocated into the rhizosphere via rhizodeposition and then interact with root hairs, soil particles, and microorganisms to form rhizosheaths. Such accumulation is particularly important as the increased SOC content in rhizosheath soils can restructure the soil system and enhance soil fertility, ultimately improving agricultural production and ensuring food security. Considering the important role of soil aggregation and stability in soil water retention and soil carbon sequestration (Bronick and Lal, 2005), the enhanced soil adhesiveness and aggregation in rhizosheaths may further suggest their potential to promote carbon sequestration and mitigating the increasing atmospheric carbon dioxide in the context of global climate change (Ndour et al., 2020).

Notably, studies are showing that the manner of SOM decomposition (rhizosphere priming) is also significantly altered by rhizosheaths (Pausch et al., 2016; Pang et al., 2017). As considerable rhizodeposits and their associated microbial metabolites are released into the soil, the substantial “new carbon” input would initiate the decomposition of “old carbon” in the soil due to the rhizosphere priming effects (Ndour et al., 2022). Although some studies have reported the intensive flux of organic carbon from the RF roots to the soil (Mo et al., 2022a; Tian et al., 2022), specific organic components (e.g., alkyl C derived from cutin, suberin, lignin, lipids, and amino acid side chains) showed a higher degree of decomposition in rhizosheath soils than in bulk soils (Mo et al., 2022a). There are also studies suggesting a significantly higher abundance of functional genes related to carbon and nitrogen cycles (including the processes of CO₂ fixation, nitrification, and denitrification) in rhizosheath soils than that in bulk soils (Wang et al., 2022). This suggests facilitated SOM dynamics and an active degradation process of organic compounds in the presence of rhizosheaths (Moreno-Espíndola et al., 2018).

Of note, rhizosheath, as both a carbon reservoir and an accelerant of carbon decomposition (Marasco et al., 2022a), plays an important role in the biogeochemical processes of SOM and tends to result in a relatively faster turnover of SOC. The proposed potential of rhizosheaths in future agriculture development is partially based on the postulations that the above-ground cover of RF plants may lead to greater soil carbon sequestration so as to improve soil fertility and mitigate CO₂ increase (Ndour et al., 2020). Nevertheless, the following aspects remain to be examined and clarified: (1) whether the formation and development of rhizosheaths have significant effects on the overall carbon budget; (2) whether the formation and development of rhizosheaths have positive or negative effects on the overall carbon budget; and (3) whether the effects of rhizosheaths on SOC turnover last for a long time and can be nonnegligible enough to be considered as a strategy for sustainable development.

5.4. Stimulation of soil microbial activity

It is widely acknowledged that the functions of rhizosheaths are closely related to soil microbial community and activity. The high microbial biomass, enzymatic activity, microbial density, and microbial community stability in rhizosheath soils provide a more favorable ecological niche for the soil microbial communities and form a hotspot for microbial colonization (Marasco et al., 2022b). Previous studies suggested that rhizosheath microbial communities are mainly dominated by *Actinobacteria*, *Alphaproteobacteria*, and saprophytic *Ascomycota* fungi (Marasco et al., 2022c). Liu et al. (2021) also found that rhizosheaths could selectively enrich specific phyla of bacteria (e.g., *Cyanobacteria* and *Bacteroidetes*) while selectively depleting others (e.g., *Proteobacteria* and *Firmicutes*). The selected microbes by rhizosheaths are able to survive in the habitats of RF plants, and thus are functionally equipped to withstand environmental stresses and support plant growth. As such, the unique microbial communities in rhizosheath soils are considered essential building blocks for rhizosheath functioning. Further studies are needed to unravel how these specific microbial communities work for the environmental functions of rhizosheaths.

Moreover, microbial activity in rhizosheath soils is stimulated by enriched resources. In addition to the high enzymatic activities (Fig. 4), studies also show that rhizosheaths have higher abundance and greater diversity of functional genes, which contribute to the cycles of carbon, nitrogen, phosphorus, and sulfur (Wang et al., 2022). Inevitably, the abundant and active microorganisms have to compete intensively to colonize favorable ecological niches. High enzymatic activities can promote microbial decomposition of macromolecular compounds, such as lignin, cellulose, and tannin in the rhizosphere, thereby accelerating the release of nutrients. This was supported by the enzymatic activities in our integration analysis (Fig. 4). A high abundance and activity of soil microorganisms can also contribute to good stability and buffering capacity, which may help plants cope with external stresses and construct a healthy soil system (Marasco et al., 2022c; Maron et al., 2018). Notably, although it is currently well-known that rhizosheaths have positive effects on microbial activities, microbial features and mechanisms in rhizosheaths remain the most complex questions to be elucidated. Answering these questions may open an important horizon for the manipulation and application of rhizosheath functions to practical situations.

6. Conclusions and prospects

Rhizosheath is a ubiquitous root-soil interface trait occurring across various plant species throughout angiosperms and exhibits critical environmental functions in multiple aspects. Ever since rhizosheaths were observed in 1887, associated studies over a hundred years demonstrated that the formation of rhizosheaths was mainly regulated by root hairs and mucilage. As rhizosheaths represent strong root-soil-microbe interactions, a comprehensive understanding of rhizosheaths

can aid in the study of the effects of strong root-soil-microbe interactions on plant growth, soil system, and holistic ecosystem. In this review, the special rhizosheath structure is introduced; the reconsideration of terminologies and methodologies is involved; the current understanding of rhizosheath formation and its influencing factors is summarized; quantitative comparisons between the features of rhizosheath soils and other soils are presented; and environmental functions of rhizosheaths are systematically dissected.

Currently, although an increasing number of studies have demonstrated the peculiarity of rhizosheaths, further in-depth research into the following aspects is urgently needed:

- (1) Precise terminology for the definition of rhizosheath soils (inner rhizosphere soils), outer rhizosphere soils, loose rhizosphere soils, and bulk soils should be standardized to establish a more cohesive literature.
- (2) Multiple effects of physical, chemical, and biological factors that manipulate rhizosheath formation and development should be explored to specify whether rhizosheaths are more genetic- or environmental-controlled. If rhizosheath acts as an adaptive strategy for plants, it is worth exploring the variations of rhizosheath traits along different environmental gradients (e.g., soil water content, soil fertility, different types of mycorrhizal colonization); which dimension of plant economic strategy does rhizosheath associate with is also an interesting issue.
- (3) Further field experiments need to be conducted to verify the postulated impacts of rhizosheaths on soil architecture, resource capture, carbon dynamics, and microbial communities, and whether these impacts are long-term maintained. The potential impact of rhizosheaths on crop breeding, CO₂ mitigation, and carbon sequestration needs careful reconsideration as rhizosheaths exhibit double-edged effects on SOC.
- (4) More work is required on broadening the number of investigated species and expanding the study sites to various ecosystems to unravel the differences in rhizosheath traits across species. Current studies are mostly based on cropland and grassland ecosystems. Speculations notwithstanding, whether rhizosheaths also occur in woody species and whether the features and functions of rhizosheaths differ across different ecosystems are still issues requiring further investigations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [grant numbers 42122054 and 42192513], Key Platform and Scientific Research Projects of Guangdong Provincial Education Department [grant numbers 2019KZDXM028 and 2020KCXTD006], Guangdong Basic and Applied Basic Research Foundation [grant number 2021B1515020082], and the Science and Technology Development Fund Project of Shenzhen [grant numbers JCYJ20190809142611503 and JCYJ20190809162205531].

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2023.116500>.

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